Business Use

Computer Forensics and cyber crime analysis

Introduction to the Course

Prof. Giuseppe Vaciago

Business Use

Overview of the Course

1. *Introduction to the Course*
2. *Cybercrime Convention*
3. *Garlasco case and the implication*
4. *IoT and Digital Forensics*
5. *Digital Forensics and Territorial Principle*
6. *Remote Forensics: Hacking Team Case*
7. *Digital Forensics and Artificial Intelligence*
8. *Case studies and practical applications*

WHO I AM?

Giuseppe Vaciago

Giuseppe Vaciago
Partner at R&P Legal
Lawyer at Milan and Professor of IT Law

Giuseppe Vaciago has been a lawyer of the Milan Bar since 2002 and for the last 10 years his primary focus has been IT Law with a focus on cyber crime. He has assisted many national and international IT companies. He is the author of many publications on cybercrime, including both scientific journals and textbooks, which have been adopted by the University where he teaches. Academically, he received his PhD on Digital Forensics from Università di Milano and he is a lecturer at Insubria University (Varese and Como) where he holds a course on IT law. He has also delivered many lectures and presentations in both Italy and abroad.

He attended Fordham Law School and Stanford Law School as a Visiting Scholar to expand his studies in his own particular research area

He is member of the executive committee of Tech and Law Center and fellow at the Nexa Center and at the Cybercrime Institute of Koln.

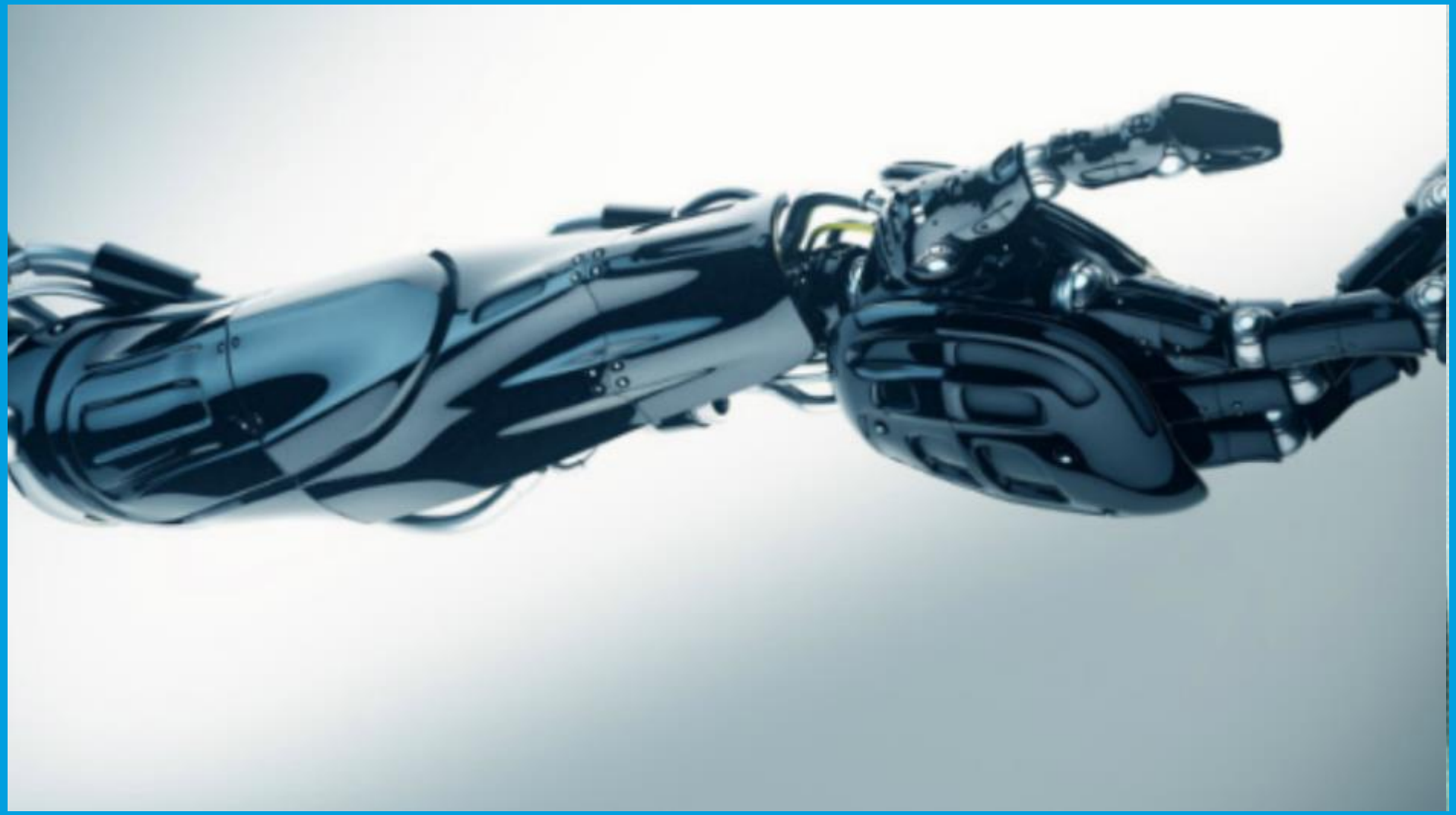
Twitter: @giuseppevaciago

Where We Are

*Technology **Between** Us*

*Technology **About** Us*

*Technology **In** Us*



A math exercise

Three plus Four,
Multiplied per Two

3+4X2=

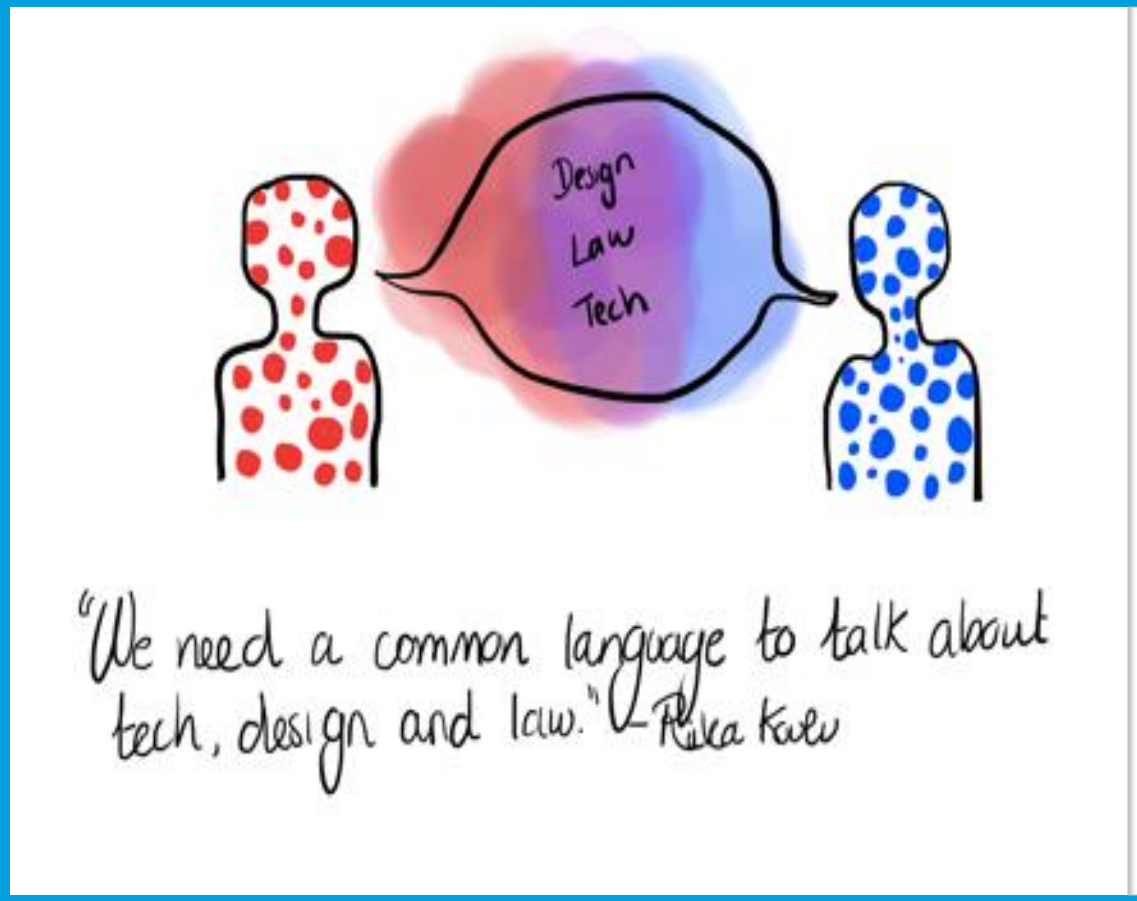
14

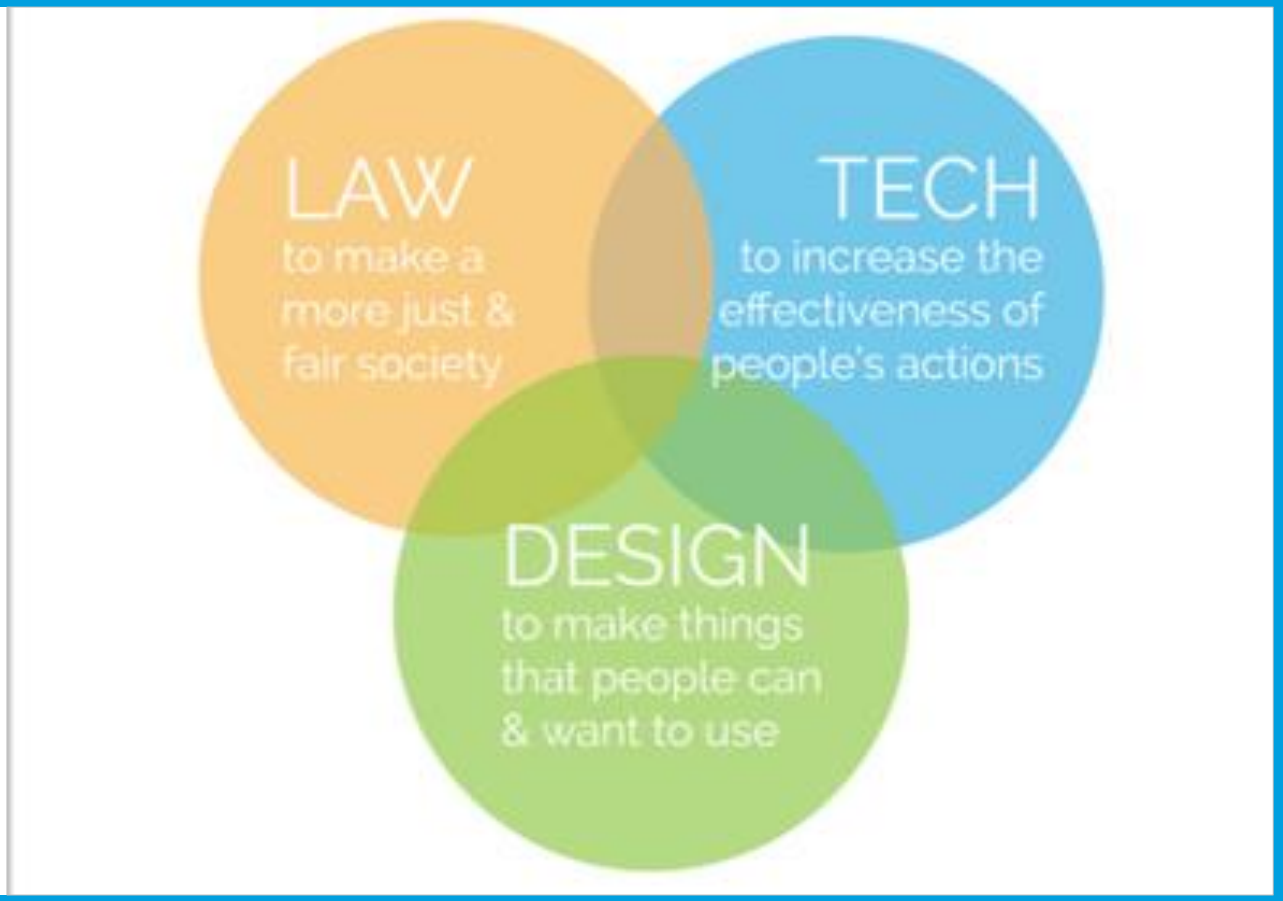
11

If the difference between 14 and 11 is important, then the natural language is not the best way to solve legal problem.

Legal Design

What is Legal Design?





® Margaret Hagan

Methodology of Legal Design

Problem 1

Information overload

- Discover the status quo
- Focus on a person
- Make sense and prioritize
- Protype
- Testing

Problem 2

Written by lawyers for lawyers

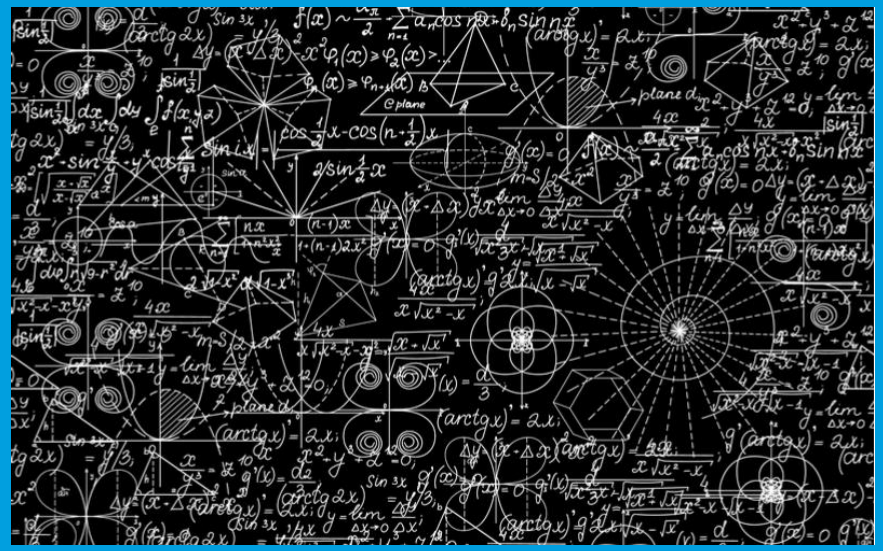
- Empathy
- Awareness of Unconscious Bias
- Respect for Community
- Trasparency About Intentions
- Commitment and Follow-Through

Dark Patterns

A dark pattern is "a user interface that has been carefully crafted to trick users into doing things, such as buying insurance with their purchase or signing up for recurring bills (Harry Brignull)

Suggestion no. 1

We must never be too complicated.



if we want to be, we would be perfect for working in Amazon Legal Department

GDPR and Algorithm Bias

Art. 22 of GDPR

1. The data subject shall have the right not to be subject to a decision **based solely on automated processing**, **including profiling**, which produces legal effects concerning him or her or similarly significantly affects him or her.

2. Paragraph 1 shall not apply if the decision:

(a) is necessary for entering into, or performance of, a contract between the data subject and a data controller;

(b) is authorised by Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights and freedoms and legitimate interests; or

(c) is based on the data subject's explicit consent.

Practical Examples

Lex Machina

Legal tasks that are being automated are largely those that used to be performed by new law school graduates and paralegals. The use of AI in areas such as contract review and management is making contracting faster and easier and lets lawyers eliminate time-consuming and repetitive tasks from their daily routines. With Prediction and litigation technology you can enter a judge's name to predict the outcome of any judgement.

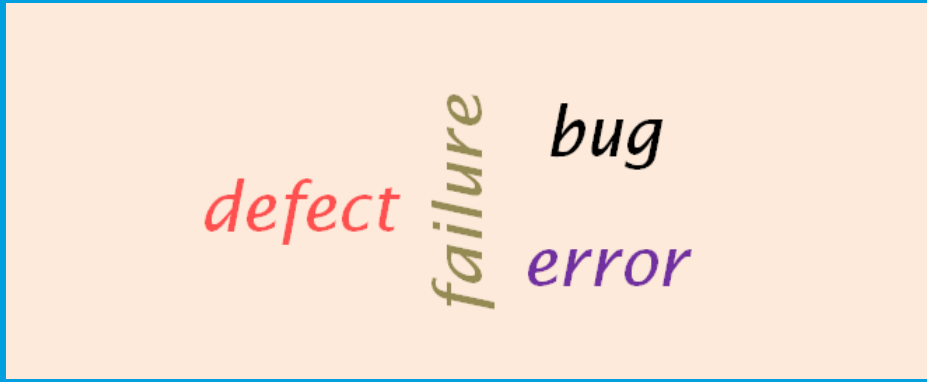
Practical Examples

Compas

COMPAS system (Correctional Offender Management Profiling for Alternative Sanctions). Nationwide, many jurisdictions use statistical algorithms to assess the likelihood a defendant will fail to appear at trial or commit a future violent crime. Propublica analysis found that white defendants who re-offended within the next two years were mistakenly labeled low risk almost twice as often as black re-offenders (48 percent vs. 28 percent).

Suggestion no. 2

Let's remember that a false positive in digital forensics can change people's lives.



Challenges: Questions?

- How can digital forensics balance the need for evidence collection with the protection of privacy rights?
- When does the use of digital forensics in workplace investigations cross ethical boundaries, such as employee surveillance?
- How should digital forensics practitioners handle potential biases in interpreting digital evidence?
- What are the ethical concerns around using data from cloud storage or third-party platforms in investigations?
- How should digital forensics be employed in investigating cyberattacks during conflicts, ensuring that evidence is collected ethically and in compliance with international law?